Modeling Databases

Module Outcome:

- Draw entities and relationships before CREATE TABLE
- Pick a primary key and a foreign key so orders know their customer
- Normalise enough that AcmeHR does not store the same phone three times

Core Concepts of ER Model

Entity

Real world object/concepts are called **entities** in ER Model.

Examples: John, Emma, Apple, Google

Attributes of an Entity

Properties of real world objects/concepts are represented as **attributes** of an entity in ER Model.

Examples:

1. name: John, age: 29
2. name: Emma, age: 25

Key Attribute

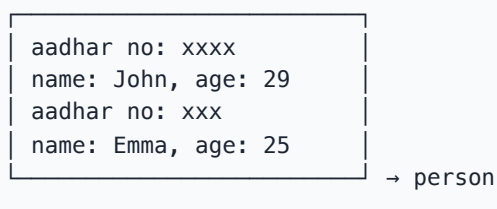
The attribute that uniquely identifies each entity is called **key attribute**.

Example: aadhar no: xxx, age: 29, name: John

Entity Type

Entity Type is a collection of entities that have the same attributes (not values).

Example:



Relationships

Association among the entities is called a **relationship**.

Examples:

- Person has a passport
- Person can have many cars
- Each student can register for many courses, and a course can have many students

Types of Relationships

1. One-to-One Relationship

An entity is related to only one entity, and vice versa.

- A person can have only one passport
- Similarly, a passport belongs to one and only one person

2. One-to-Many Relationship

An entity is related to many other entities.

- A person can have many cars. But a car belongs to only one person

3. Many-to-Many Relationship

Multiple entities are related to multiple entities.

- Each student can register to multiple courses
- Similarly each course is taken by multiple students

Cardinality Ratio

Cardinality in DBMS defines the maximum number of times an instance in one entity can relate to instances of another entity.

- one-to-one (1:1)
- one-to-Many (1:m)
- Many-to-one (m:1)
- Many-to-Many (m:n)

Applying ER Model Concepts — E-Commerce Application

In a typical e-commerce application:

- Customer has only one cart. A cart belongs to only one customer.
- Customer can add products to cart.
- Cart contains multiple products.
- Customer can save multiple addresses in the application.

Entity types: Customer, Product, Cart, Address

Relationships:

Relationship	Type
Customer ↔ Cart	One-to-One
Cart ↔ Products	Many-to-Many
Customer ↔ Address	One-to-Many

Attributes:

Customer	Product	Address	Cart
id (PK)	product id (PK)	id (PK)	cart id (PK)
name	price	pin code	total price
age	name	door no	
	brand	city	
	category		

ER Model of E-Commerce Application:

```
Customer --1-- has --1-- Cart
Customer --1-- has --n-- Address
Cart --m-- contains --n-- Product
```

ER Model to Relational Database Mapping:

- Entity Types → Tables
- Attributes → Columns
- Key Attribute → Primary Key

Primary Keys

A minimal set of attributes (columns) in a table that uniquely identifies rows in a table.

Table	Primary Key	Columns
customer	id	id, name, age
address	id	id, pin_code, door_no, city
cart	id	id, total_price
product	id	id, name, price, brand, category

Foreign Keys

Relationship Between Customer and Address — One to Many:

We store the primary key of a customer in the address table. This new column in the table that refers to the primary key of another table is called **Foreign Key**.

address table
id (PK), pin_code, door_no, ..., customer_id (FK)

Here, `customer_id` is the foreign key that stores `id` (primary key) of customer.

Relationship Between Cart and Customer — One to One:

Similar to one-to-many relationship, but we need to ensure that only one cart is associated to a customer.

cart table
id (PK), total_price, customer_id (FK, UNIQUE)

Relationship Between Cart and Products — Many to Many:

We cannot store either the primary key of a product in the cart table or vice versa. To store the relationship, we use a **Junction Table** (`cart_product`).

cart	cart_product	product
id: 1, 2	cart_id (FK), product_id (FK)	id, name, ...

Properties related to a many-to-many relationship (e.g., quantity of each product in the cart) should be stored in the junction table.

Relational Database Schema:

customer	address	cart	product	cart_product
id, name, age	id, pincode, door_no, city, customer_id	id, total_price, customer_id	id, name, price, brand, category	id, cart_id, product_id, quantity

Creating a Relational Database

Table Constraints

When creating tables, we apply constraints to enforce data integrity:

Constraint	Purpose	Example
`NOT NULL`	Column cannot be empty	`name VARCHAR(250) NOT NULL`
`UNIQUE`	No duplicate values allowed	`email VARCHAR(250) UNIQUE`
`DEFAULT`	Value used when none is provided	`status VARCHAR(20) DEFAULT 'active'`
`CHECK`	Validates a condition	`price INT CHECK (price > 0)`
`PRIMARY KEY`	Unique row identifier	`id INTEGER PRIMARY KEY`
`FOREIGN KEY`	References another table	`customer_id REFERENCES customer (id)`

Example: Enforce one cart per customer using `UNIQUE` on the foreign key.

```
CREATE TABLE cart (
    id INT NOT NULL PRIMARY KEY AUTO_INCREMENT,
    customer_id INTEGER NOT NULL UNIQUE,
    total_price INTEGER,
    FOREIGN KEY(customer_id) REFERENCES customer(id) ON DELETE CASCADE
);
```

`UNIQUE` on `customer_id` ensures the one-to-one relationship between customer and cart.

Primary Key Syntax

```
CREATE TABLE table_name (
    C1 t1 NOT NULL PRIMARY KEY,
    ...
    Cn tn
);
```

Foreign Key Syntax

```
CREATE TABLE table2 (
    C1 t1 NOT NULL PRIMARY KEY,
    C2 t2,
    FOREIGN KEY (C2) REFERENCES table1 (C3)
    ON DELETE CASCADE
);
```

`FOREIGN KEY (C2) REFERENCES table1 (C3)` ensures that foreign key can only contain values that are in the referenced primary key.

ON DELETE CASCADE: If a row in table1 is deleted, then all its related rows in table2 will also be deleted.

> **Note:** In MySQL, foreign key constraints are enforced automatically when using the **InnoDB** storage engine (the default). Ensure referenced tables use InnoDB: `ENGINE=InnoDB`.

Creating Tables

```
CREATE TABLE customer (
    id INT NOT NULL PRIMARY KEY AUTO_INCREMENT,
    name VARCHAR(250),
    age INT
);

CREATE TABLE product (
    id INT NOT NULL PRIMARY KEY AUTO_INCREMENT,
    name VARCHAR(250),
    price INT,
    brand VARCHAR(250),
    category VARCHAR(250)
);

CREATE TABLE address (
    id INT NOT NULL PRIMARY KEY AUTO_INCREMENT,
    pin_code INTEGER,
    door_no VARCHAR(250),
    city VARCHAR(250),
    customer_id INTEGER,
    FOREIGN KEY(customer_id) REFERENCES Customer(id) ON DELETE CASCADE
);

CREATE TABLE cart (
    id INT NOT NULL PRIMARY KEY AUTO_INCREMENT,
    customer_id INTEGER NOT NULL UNIQUE,
```

```
    total_price INTEGER,
    FOREIGN KEY(customer_id) REFERENCES customer(id) ON DELETE CASCADE
);

CREATE TABLE cart_product (
    id INT NOT NULL PRIMARY KEY AUTO_INCREMENT,
    cart_id INTEGER,
    product_id INTEGER,
    quantity INTEGER,
    FOREIGN KEY(cart_id) REFERENCES cart(id) ON DELETE CASCADE,
    FOREIGN KEY(product_id) REFERENCES product(id) ON DELETE CASCADE
);
```